

Concentration Inequalities

Markov, Chebyshev, and Chernoff Bounds

Hoa Vu
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Concentration Inequalities

- ▶ In big-data systems: sampling, hashing, and noise.
- ▶ **How far can we deviate from expectation?**
- ▶ Better assumptions $\Rightarrow$ stronger bounds.

Markov's Inequality

Assumption: $X \geq 0$ and $\mathbb{E}[X] < \infty$.

$$\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}, \quad a > 0$$

- ▶ Uses only the mean.
- ▶ Very general, often loose.

Proof of Markov's Inequality

For $X \geq 0$ and $a > 0$, let $I = \mathbf{1}\{X \geq a\}$.

$$\mathbb{E}[X] \geq \mathbb{E}[XI] \geq \mathbb{E}[aI] = a \Pr(X \geq a)$$

Divide both sides by a :

$$\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Markov Example

Suppose X = number of delayed jobs in one minute.

$$\mathbb{E}[X] = 4, \quad a = 20$$

$$\Pr(X \geq 20) \leq \frac{4}{20} = 0.2$$

Interpretation: at most 20% of minutes have 20 or more delayed jobs.

Variance Refresher

Let $\mu = \mathbb{E}[X]$.

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2]$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

- ▶ Measures spread around the mean.
- ▶ Needed by Chebyshev's inequality.

Variance of a Bernoulli Random Variable

Let $X \sim \text{Bernoulli}(p)$, so $X \in \{0, 1\}$.

$$\mathbb{E}[X] = p, \quad X^2 = X \Rightarrow \mathbb{E}[X^2] = p$$

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = p - p^2 = p(1 - p)$$

Example: if $p = 0.2$, then

$$\text{Var}(X) = 0.2 \cdot 0.8 = 0.16.$$

Averaging i.i.d. Samples Reduces Variance

Let $X_1, \dots, X_n$ be i.i.d. with mean μ and variance σ^2 . Define

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Using independence:

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) = n\sigma^2.$$

$$\text{Var}(\bar{X}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}.$$

Example: Bernoulli Average

If $X_i \sim \text{Bernoulli}(p)$, then $\text{Var}(X_i) = p(1 - p)$.

For the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$:

$$\text{Var}(\bar{X}) = \frac{p(1 - p)}{n}.$$

Example with $p = 0.2$:

$$\text{Var}(X_i) = 0.16, \quad \text{Var}(\bar{X}) = \frac{0.16}{n}.$$

At $n = 100$, $\text{Var}(\bar{X}) = 0.0016$, much smaller than 0.16.

Chebyshev's Inequality

Assumption: finite mean μ and variance σ^2 .

$$\Pr(|X - \mu| \geq t\sigma) \leq \frac{1}{t^2}, \quad t > 0$$

- ▶ Uses mean and variance.
- ▶ Two-sided deviation control.

Proof of Chebyshev's Inequality

Let $Y = (X - \mu)^2 \geq 0$, where $\mu = \mathbb{E}[X]$ and $\sigma^2 = \text{Var}(X)$.

Apply Markov to Y with threshold $t^2\sigma^2$:

$$\Pr(Y \geq t^2\sigma^2) \leq \frac{\mathbb{E}[Y]}{t^2\sigma^2}$$

$$\Pr(|X - \mu| \geq t\sigma) = \Pr((X - \mu)^2 \geq t^2\sigma^2) \leq \frac{\text{Var}(X)}{t^2\sigma^2} = \frac{1}{t^2}.$$

Chebyshev Example

Suppose query latency X has

$$\mu = 100 \text{ ms}, \quad \sigma = 20 \text{ ms}$$

$$\Pr(X \geq 160) \leq \Pr(|X - 100| \geq 60) \leq \frac{20^2}{60^2} = \frac{1}{9} \approx 0.111$$

So the chance of latency at least 160 ms is at most 11.1%.

Problem: Estimating a Biased Coin

Setup: A coin lands heads with unknown probability p .

We flip it n times and observe $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$.

Goal: use the sample mean

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$$

to estimate p to within **additive error** ε :

$$\Pr(|\hat{p} - p| \geq \varepsilon) \leq \delta.$$

Question: How large must n be?

Applying Chebyshev to Coin Estimation

We know $\mathbb{E}[\hat{p}] = p$ and

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \leq \frac{1}{4n},$$

since $p(1-p) \leq \frac{1}{4}$ for all $p \in [0, 1]$.

Apply Chebyshev with threshold ε :

$$\Pr(|\hat{p} - p| \geq \varepsilon) \leq \frac{\text{Var}(\hat{p})}{\varepsilon^2} \leq \frac{1}{4n\varepsilon^2}.$$

Sample Complexity via Chebyshev

We want the failure probability to be at most δ :

$$\frac{1}{4n\epsilon^2} \leq \delta \implies n \geq \frac{1}{4\delta\epsilon^2}.$$

Example: $\epsilon = 0.05$, $\delta = 0.05$ (95% confidence, $\pm 5\%$ error):

$$n \geq \frac{1}{4 \cdot 0.05 \cdot (0.05)^2} = \frac{1}{0.0005} = 2000.$$

With 2000 flips, our estimate is within 0.05 of the true p with probability at least 95%.

Chernoff Bounds (Bernoulli Sum)

Let $X = \sum_{i=1}^n X_i$, where $X_i \in \{0, 1\}$ are independent, $\mu = \mathbb{E}[X]$.

For $0 < \delta \leq 1$:

$$\Pr(X \geq (1 + \delta)\mu) \leq e^{-\mu\delta^2/3}$$

$$\Pr(X \leq (1 - \delta)\mu) \leq e^{-\mu\delta^2/2}$$

Stronger assumptions $\Rightarrow$ exponentially small tails.

Chernoff Example

X = number of bad packets in $n = 10,000$ packets, with bad probability $p = 0.01$.

$$\mu = np = 100, \quad \delta = 0.5$$

$$\Pr(X \geq 150) = \Pr(X \geq (1 + 0.5)\mu) \leq e^{-100 \cdot (0.5)^2/3} = e^{-8.33} \approx 2.4 \times 10^{-4}$$

A 50% overshoot is very unlikely.

Coin Estimation via Chernoff

Same setup: $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$, estimator $\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i$.

Let $S = n\hat{p} = \sum_{i=1}^n X_i$, so $\mu = \mathbb{E}[S] = np$.

Convert additive error to **multiplicative** error for each tail:

$$|\hat{p} - p| \geq \varepsilon \iff |S - np| \geq n\varepsilon \iff \left| \frac{S}{\mu} - 1 \right| \geq \frac{\varepsilon}{p}.$$

Set $\delta_{\text{rel}} = \varepsilon/p$ and apply Chernoff to each tail.

Applying Chernoff to Both Tails

Upper tail:

$$\Pr(\hat{p} \geq p + \varepsilon) = \Pr(S \geq (1 + \frac{\varepsilon}{p})\mu) \leq e^{-\mu(\varepsilon/p)^2/3} = e^{-n\varepsilon^2/(3p)}.$$

Lower tail:

$$\Pr(\hat{p} \leq p - \varepsilon) = \Pr(S \leq (1 - \frac{\varepsilon}{p})\mu) \leq e^{-\mu(\varepsilon/p)^2/2} = e^{-n\varepsilon^2/(2p)}.$$

Union bound + worst case $p \leq 1 \Rightarrow e^{-n\varepsilon^2/(3p)} \leq e^{-n\varepsilon^2/3}$:

$$\Pr(|\hat{p} - p| \geq \varepsilon) \leq 2 e^{-n\varepsilon^2/3}.$$

Sample Complexity via Chernoff

Set the failure probability to at most δ :

$$2e^{-n\varepsilon^2/3} \leq \delta \implies n \geq \frac{3 \ln(2/\delta)}{\varepsilon^2}.$$

Comparison ($\varepsilon = 0.05$, $\delta = 0.001$):

Method	Required n
Chebyshev	$\frac{1}{4\delta\varepsilon^2} = 100,000$
Chernoff	$\frac{3 \ln(2000)}{0.0025} \approx 9,120$

Chernoff needs **10×** fewer samples when δ is small.